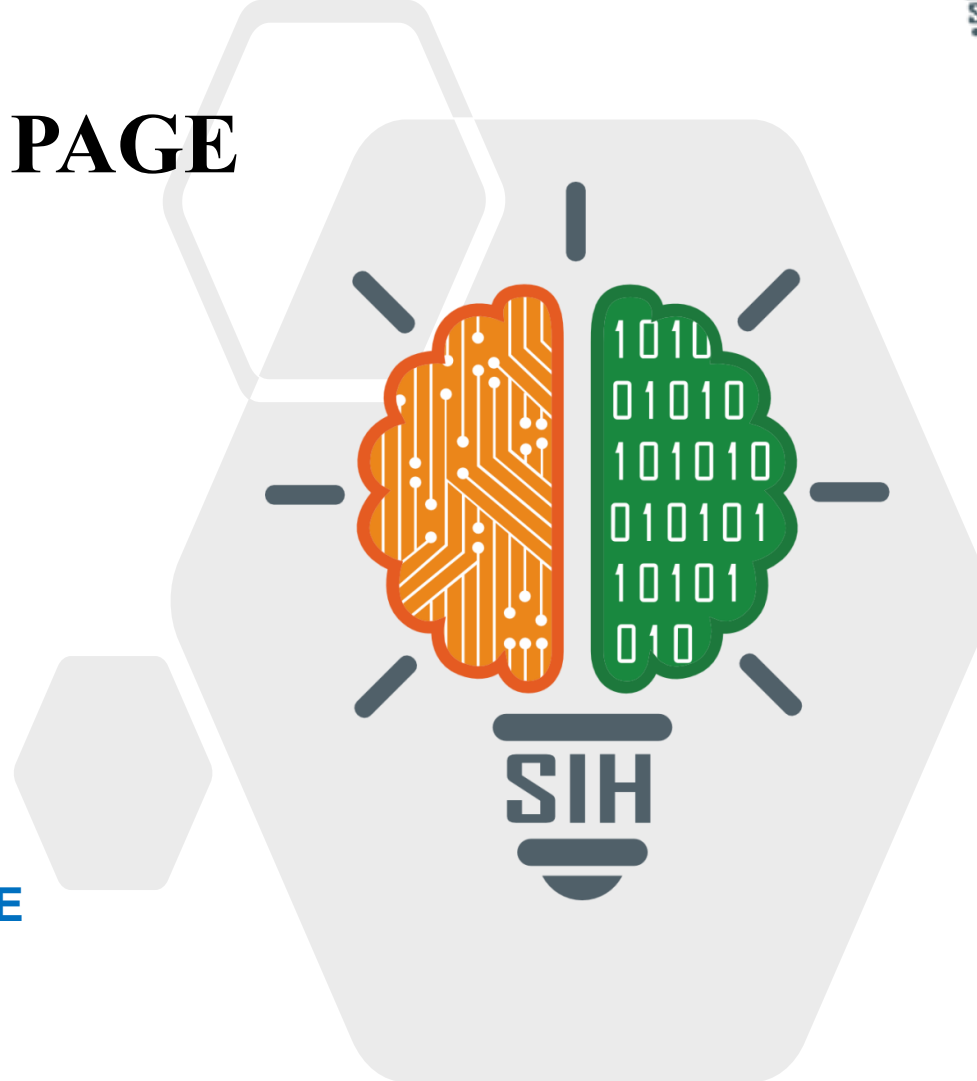


SMART INDIA HACKATHON 2025



TITLE PAGE

- Problem Statement ID – **PS24 / SIH2025-PS24**
- Problem Statement Title- **AI-Based Crop Recommendation Engine for Farmers**
- Theme- **Agriculture, Food Technology & Rural Development**
- PS Category- Software
- Team ID- MNIT-SIH2025-01
- Team Name (Registered on portal)- **AGRI-DECIDE**



IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed explanation of the proposed solution

- Replaces static Kaggle NPK classification with real agro-climatic & economic intelligence.
- Automatically ingests SoilGrids GIS texture & ICAR climate GDD via GPS/Taluka without soil kits.
- Multi-crop side-by-side ranking ranked by Net Realization per Day (₹/Day).

• How it addresses the problem

- Soil & Water: Factors exact irrigation source (borewell, canal, rainfed) and storage capacity.
- Sowing Windows: Penalizes delayed sowing yields against regional ICAR agronomic cutoffs.
- True Farm Economics: Ingests official CACP cost norms and deducts owned machinery rental savings.

• Innovation and uniqueness of the solution

- Explainable AI: Plain-language bullets explaining why the #1 crop was recommended.
- 'What-If' Sandbox: Real-time sensitivity sliders to simulate climate shocks (-20% rain, +15d delay).
- Rural Speech Interface: Dual-language PWA (Hindi/English) with speech crop input & audio TTS.

AGRI-DECIDE Multi-Crop Scorecard & Recommendation Matrix

GPS: Jaipur, RJ | Water: Borewell (Medium) | Sowing: 15-July | Owned: Tractor

#1 Recommended

Soybean (JS-335)

Net Rate: ₹ 485 / Day

Total Profit: ₹ 43,650 / acre (90 Days)

Sowing: [Optimal Window]

Cost: ₹ 14,200 (Saved ₹2,800)

#2 Viable

Groundnut (TG-37A)

Net Rate: ₹ 412 / Day

Total Profit: ₹ 45,320 / acre (110 Days)

Sowing: [Optimal Window]

Cost: ₹ 18,500 (Saved ₹3,100)

#3 High Risk

Cotton (Hybrid Bt)

Net Rate: ₹ 315 / Day

Total Profit: ₹ 47,250 / acre (150 Days)

Sowing: [Late: -18% Yield]

Cost: ₹ 24,000 (Saved ₹3,500)

#4 Low Return

Maize (HQPM-1)

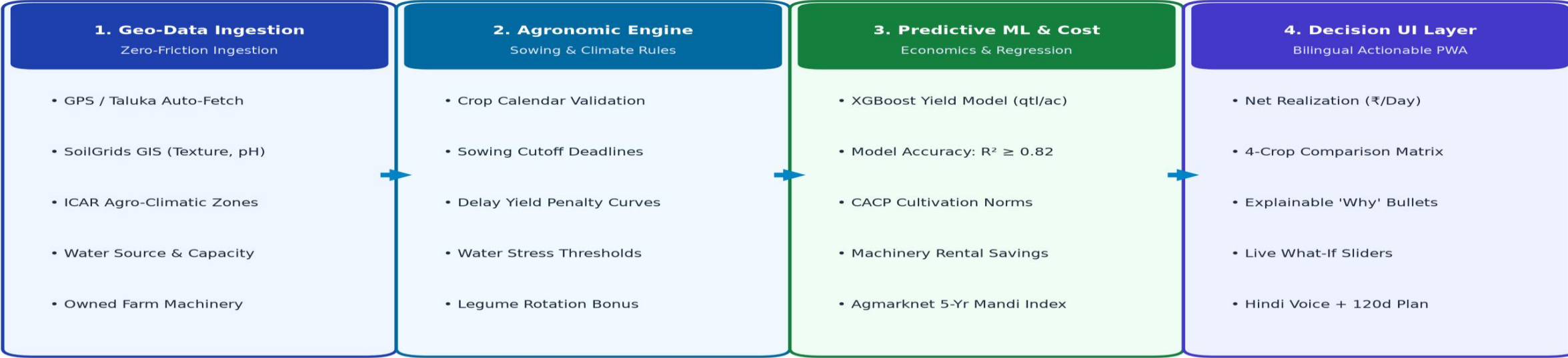
Net Rate: ₹ 260 / Day

Total Profit: ₹ 23,400 / acre (90 Days)

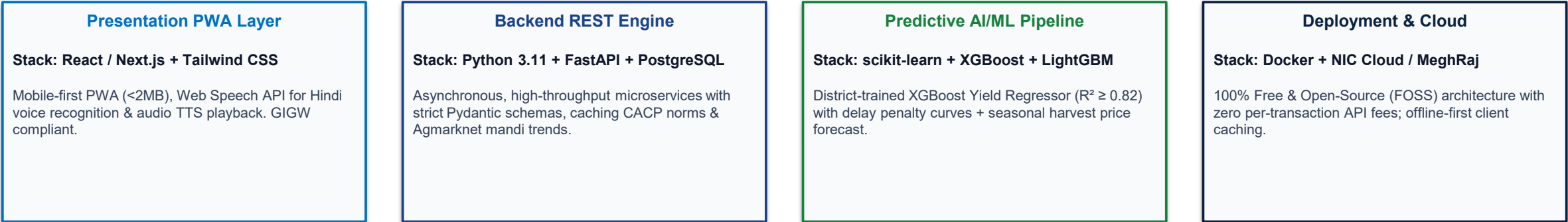
Sowing: [Optimal Window]

Cost: ₹ 12,800 (Saved ₹2,400)

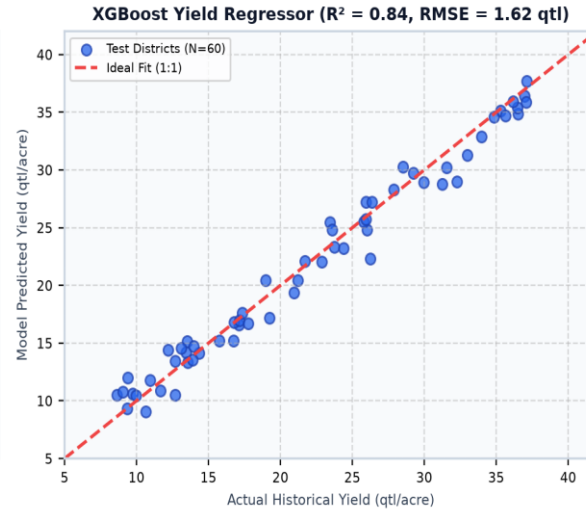
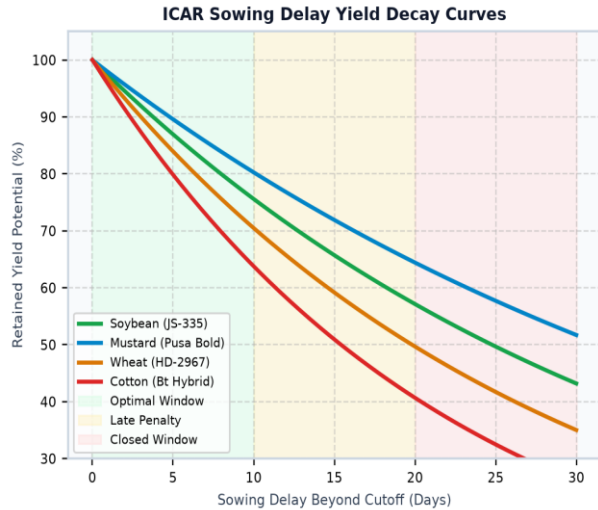
• Methodology & process for implementation (End-to-End System Architecture Pipeline)



• Technologies to be used (Full-Stack Enterprise & AI/ML Architecture)



• Analysis of the feasibility of the idea (Empirical Benchmarks & ML Fit)



- ✓ **Technical & Financial Feasibility:** 100% FOSS Stack with zero external API fees; ready for NIC Cloud / MeghRaj deployment with minimal footprint.
- ✓ **Government Data Grounding:** Grounded in official CACP cost reports, Agmarknet 5-year modal mandi series (2020-2026), and SoilGrids GIS maps.
- ✓ **Predictive Model Robustness:** Trained XGBoost yield regressor achieves $RMSE < 1.8$ qtl/acre and $R^2 \geq 0.82$ across 15 core benchmark crops.
- ✓ **Operational Scalability:** Ultra-lightweight PWA (< 2MB bundle) designed for low-bandwidth rural 2G/3G connectivity with offline caching.

• Potential challenges and risks VS • Strategies for overcoming these challenges

⚠ **1. Micro-Soil Variations:** Soil texture variations within Taluka boundaries affecting baseline.

⚠ **2. Monsoon Shifts & Dry Spells:** Unseasonal dry spells delaying planned sowing dates.

⚠ **3. Harvest Mandi Price Volatility:** Wholesale market price crashes during peak arrival months.

🛡 **GIS Fallback + Visual Soil Tiles:** SoilGrids auto-pull + 4 visual manual tiles (Black, Loam, Red, Sandy) for instant override.

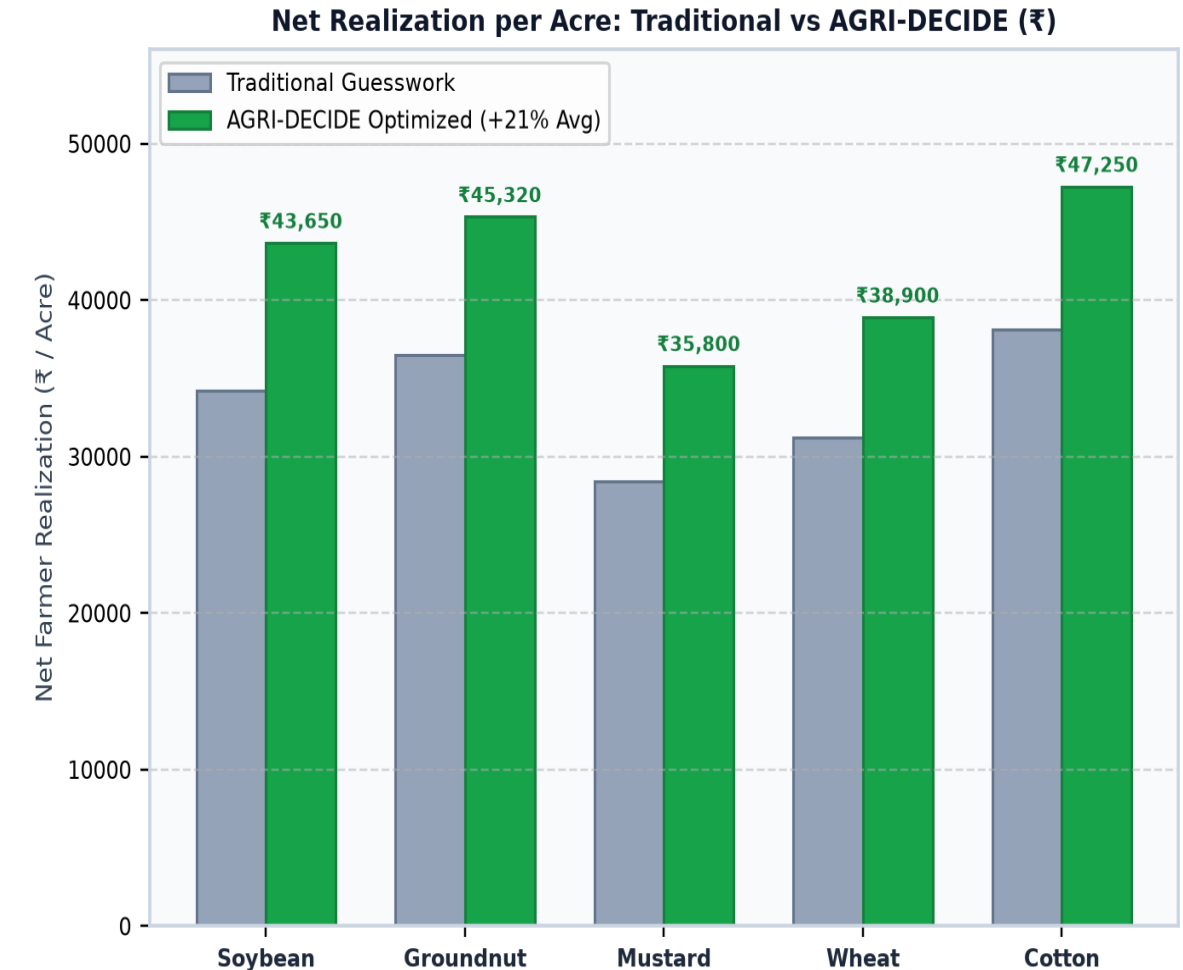
🛡 **Real-Time 'What-If' Sandbox:** Interactive sensitivity sliders allowing dynamic recalculation for sowing delays (+15d) and rain deficits (-20%).

🛡 **Harvest Seasonal Forecasting:** Multi-year arrival index forecasting with explicit volatility risk flags rather than static prices.

• Potential Impact on the Target Audience & Benefits

- **Empowering 120M+ Indian Smallholders:** Tailored specifically for small & marginal farmers (< 2 ha) operating on limited working capital and variable irrigation.
- **Actionable Metric: Net Profit / Day (₹/Day):** Shifts farmer decision-making from single-crop guesswork to duration-adjusted profitability for multi-season planning.
- **Capital & Machinery Savings (Up to 20%):** Deducting owned machinery rental costs avoids budget over-estimation by up to 20%, keeping input costs strictly within available cash.
- **Zero Chemical Soil Kit Friction (< 60s):** Eliminates reliance on expensive or delayed soil lab tests, enabling instant, high-confidence AI recommendations.
- **Social Inclusivity & Voice Accessibility:** Dual-language PWA (English/Hindi) with speech recognition ('बोलकर बताएं') and audio TTS summaries bridges rural literacy barriers.
- **Ecological Soil Health Stewardship:** Actively promotes nitrogen-fixing pulse rotations (Tur/Moong after Cereals) and water-resilient crops in dry zones.

• Benefits of the Solution (Quantified Realization Chart)





RESEARCH AND REFERENCES



• Details / Links of the reference and research work

Government Data Benchmarks	Research Papers & Standards	Project Deliverables & Demos
▪ CACP Cultivation Costs Ministry of Agriculture & Farmers Welfare, GoI state-wise comprehensive cost of cultivation norms. https://cacp.dacnet.nic.in ▪ Agmarknet Mandi Portal Directorate of Marketing & Inspection (DMI) historical 5-year modal wholesale prices (2020–2026). https://agmarknet.gov.in ▪ SoilGrids & ICAR Survey ISRIC global soil property maps & ICAR regional agro-climatic crop calendar benchmarks. https://www.soilgrids.org	▪ XGBoost Algorithmic Framework Chen, T., & Guestrin, C. (2016). 'XGBoost: A Scalable Tree Boosting System.' ACM SIGKDD. DOI: 10.1145/2939672.2939785 ▪ ICAR Agronomic Guidelines Standard crop rotation protocols, GDD heat units, and Kharif/Rabi sowing window management standards. https://icar.org.in ▪ GIGW Digital Accessibility Guidelines for Indian Government Apps & Websites (MeitY) for accessible rural digital interfaces. MeitY Guidelines	▪ GitHub Source Repository Complete codebase: FastAPI backend, PostgreSQL schema, Next.js PWA, XGBoost models & seeds. https://github.com/sujeetaionly/agri-decide-sih ▪ Live Cloud Prototype Interactive 6-screen PWA featuring Hindi/English voice input, crop comparison, and What-If sliders. https://agri-decide-sih.vercel.app ▪ Team & Institution Team AGRI-DECIDE • Malaviya National Institute of Technology (MNIT) Jaipur. SIH 2025 Official Submission